SQL PROJECT - Customer Churn Analytics & Revenue Risk Modeling in European Banking

Question Set 1 - Easy

1. What is the total size of the bank's customer base, and what percentage of customers have churned?

```
SELECT COUNT(*) FROM bank_churn_clean;
SELECT
COUNT(*) AS total_customers,
SUM(Exited) AS churned_customers,
ROUND(SUM(Exited)*100.0/COUNT(*), 2) AS churn_rate
FROM bank_churn_clean;
```

2. Do male and female customers exhibit different churn behaviors, indicating a need for gender-specific retention strategies?

```
SELECT
Gender,
COUNT(*) AS customers,
SUM(Exited) AS churned_customers,
ROUND(SUM(Exited)*100.0/COUNT(*),2) AS churn_rate
FROM bank_churn_clean
GROUP BY Gender;
```

3. How is the customer base distributed across different age segments, and which age group dominates the portfolio?

```
SELECT *,
CASE
   WHEN Age < 30 THEN 'Under 30'
   WHEN Age BETWEEN 30 AND 45 THEN '30-45'
   WHEN Age BETWEEN 46 AND 60 THEN '46-60'
   ELSE '60+'
END AS AgeGroup
FROM bank_churn_clean;
```

4. Which age groups are most prone to churn, and how does churn risk vary across lifecycle stages?

```
SELECT
AgeGroup,
COUNT(*) AS customers,
SUM(Exited) AS churned_customers,
ROUND(SUM(Exited)*100.0/COUNT(*),2) AS churn_rate
FROM customer_segments
GROUP BY AgeGroup;
```

5. How does the number of products owned by a customer influence their likelihood to churn?

```
SELECT
NumOfProducts,
COUNT(*) AS customers,
SUM(Exited) AS churned_customers,
ROUND(SUM(Exited)*100.0/COUNT(*),2) churn_rate
FROM bank_churn_clean
GROUP BY NumOfProducts
ORDER BY NumOfProducts;
```

6. What is the impact of customer engagement (active vs inactive) on churn probability?

```
SELECT
IsActiveMember,
COUNT(*) AS customers,
SUM(Exited) AS churned_customers,
ROUND(SUM(Exited)*100.0/COUNT(*),2) AS churn_rate
FROM bank_churn_clean
GROUP BY IsActiveMember;
```

Question Set 2 - Moderate

1. How does customer tenure influence churn, and are newer customers more at risk compared to long-term customers?

```
SELECT
TenureSegment,
COUNT(*) AS customers,
SUM(Exited) AS churned_customers,
ROUND(SUM(Exited)*100.0/COUNT(*),2) AS churn_rate,
ROUND(SUM(Exited) / (SELECT SUM(Exited) FROM customer_segments) * 100, 2) AS
churn_contribution
FROM customer_segments
GROUP BY TenureSegment
ORDER BY churn_rate DESC;
```

2. Which balance segments contribute the most to total churn, and where is the bank losing the most customers?

```
SELECT
BalanceSegment,
COUNT(*) AS customers,
SUM(Exited) AS churned_customers,
ROUND(SUM(Exited)*100.0/COUNT(*),2) AS churn_rate,
ROUND(SUM(Exited)/(SELECT SUM(Exited) FROM customer_segments)*100,2) AS
churn_contribution
FROM customer_segments
GROUP BY BalanceSegment
ORDER BY churn_rate DESC;
```

3. Does customer income level (salary) influence churn, and which income segment is most vulnerable?

```
SELECT
CASE
   WHEN EstimatedSalary < 50000 THEN 'Low Salary'
   WHEN EstimatedSalary BETWEEN 50000 AND 100000 THEN 'Medium Salary'
   ELSE 'High Salary'
END AS SalarySegment,
COUNT(*) AS customers,
SUM(Exited) AS churned_customers,
ROUND(SUM(Exited)*100.0/COUNT(*),2) AS churn_rate
FROM bank_churn_clean
GROUP BY SalarySegment
ORDER BY churn_rate DESC;
```

4. How does creditworthiness impact churn behavior, and are low-credit customers more likely to leave?

```
SELECT
CreditScoreGroup,
COUNT(*) AS customers,
SUM(Exited) AS churned_customers,
ROUND(SUM(Exited)*100.0/COUNT(*),2) AS churn_rate
FROM customer_segments
GROUP BY CreditScoreGroup
ORDER BY churn_rate DESC;
```

5. Which geographic regions contribute the most to total churn, and where should retention efforts be prioritized?

```
SELECT
Geography,
COUNT(*) AS customers,
SUM(Exited) AS churned_customers,
ROUND(SUM(Exited)*100.0/COUNT(*),2) AS churn_rate,
ROUND(SUM(Exited)/(SELECT SUM(Exited) FROM bank_churn_clean)*100,2) AS churn_contribution
FROM bank_churn_clean
GROUP BY Geography
ORDER BY churn_rate DESC;
```

6. How does churn behavior vary across different age groups within each geography?

```
SELECT
Geography,
AgeGroup,
COUNT(*) AS customers,
SUM(Exited) AS churned_customers,
ROUND(SUM(Exited)*100.0/COUNT(*),2) AS churn_rate
FROM customer_segments
GROUP BY Geography, AgeGroup
ORDER BY Geography;
```

7. Which product usage category shows the highest churn risk, and does product diversification improve retention?

```
SELECT
CASE
   WHEN NumOfProducts = 1 THEN 'A'
   WHEN NumOfProducts = 2 THEN 'B'
   WHEN NumOfProducts = 3 THEN 'C'
   WHEN NumOfProducts = 4 THEN 'D'
END AS Product_Category,
COUNT(*) AS total_customers,
SUM(CASE WHEN Exited = 1 THEN 1 ELSE 0 END) AS churned_customers,
SUM(CASE WHEN Exited = 0 THEN 1 ELSE 0 END) AS active_customers,
ROUND(AVG(Exited)*100,2) AS churn_rate
FROM bank_churn_clean
GROUP BY Product_Category
ORDER BY Product_Category;
```

8. How does customer activity influence churn across different balance segments?

```
SELECT
BalanceSegment,
IsActiveMember,
COUNT(*) AS customers,
SUM(Exited) AS churned_customers,
ROUND(SUM(Exited)*100.0/COUNT(*),2) AS churn_rate
FROM customer_segments
GROUP BY BalanceSegment, IsActiveMember
ORDER BY churn_rate DESC;
```

9. What are the key differences in financial and demographic profiles between churned and retained customers?

```
SELECT
Exited,
COUNT(*) AS customers,
ROUND(AVG(Age),2) AS avg_age,
ROUND(AVG(CreditScore),2) AS avg_credit_score,
ROUND(AVG(Balance),2) AS avg_balance,
ROUND(AVG(EstimatedSalary),2) AS avg_salary,
ROUND(AVG(Tenure),2) AS avg_tenure
FROM customer_segments
GROUP BY Exited;
```

Question Set 3 – Advanced

1. Which customer segments pose the highest financial risk due to churn in terms of balance loss?

```
SELECT
BalanceSegment,
SUM(Balance) AS total_balance,
SUM(CASE WHEN Exited = 1 THEN Balance ELSE 0 END) AS balance_lost
FROM customer_segments
GROUP BY BalanceSegment;
```

2. What is the total financial impact of churn in terms of lost customer balance?

```
SELECT
SUM(Balance) AS total_balance_lost
FROM bank_churn_clean
WHERE Exited = 1;
```

3. Which geographic region contributes the highest revenue loss due to churn, and what is the total absolute loss (in €) for each region?

```
SELECT
Geography,
SUM(Balance) AS total_balance,
SUM(CASE WHEN Exited = 1 THEN Balance ELSE 0 END) AS balance_lost,
FROM bank_churn_clean
GROUP BY Geography
ORDER BY balance_lost DESC;
```

4. What percentage of total customer balance is lost due to churn in each region?

```
SELECT
Geography,
SUM(Balance) AS total_balance,
SUM(CASE WHEN Exited = 1 THEN Balance ELSE 0 END) AS balance_lost,
ROUND(SUM(CASE WHEN Exited = 1 THEN Balance ELSE 0 END) / SUM(Balance) * 100, 2) AS
loss_percentage
FROM bank_churn_clean
GROUP BY Geography
ORDER BY balance_lost DESC;
```

5. Are high-value customers (balance > 100K) more likely to churn, posing a higher revenue risk?

```
SELECT
COUNT(*) AS high_value_customers,
SUM(Exited) AS churned_high_value,
ROUND(SUM(Exited)*100.0/COUNT(*),2) AS churn_rate
FROM bank_churn_clean
WHERE Balance > 100000;
```

6. Which combination of high balance and low activity represents the highest churn risk segment?

```
SELECT
BalanceSegment,
IsActiveMember,
COUNT(*) AS customers,
SUM(Exited) AS churned_customers,
ROUND(SUM(Exited)*100.0/COUNT(*),2) AS churn_rate
FROM customer_segments
GROUP BY BalanceSegment, IsActiveMember
ORDER BY churn_rate DESC;
```

7. How does churn behavior among high-value customers vary across different geographies?

```
SELECT
Geography,
BalanceSegment,
COUNT(*) AS customers,
SUM(Exited) AS churned_customers,
ROUND(SUM(Exited)*100.0/COUNT(*),2) AS churn_rate
FROM customer_segments
WHERE BalanceSegment = 'High Balance'
GROUP BY Geography, BalanceSegment
ORDER BY churn_rate DESC;
```

8. What is the relationship between customer engagement and financial value in driving churn risk?

```
SELECT
BalanceSegment,
IsActiveMember,
COUNT(*) AS customers,
SUM(Exited) AS churned_customers,
ROUND(SUM(Exited)*100.0/COUNT(*),2) AS churn_rate
FROM customer_segments
GROUP BY BalanceSegment, IsActiveMember
ORDER BY churn_rate DESC;
```

9. Which regions present the highest overall risk when combining churn rate and revenue loss (Geographic Risk Index)?

```
SELECT Geography,
SUM(Balance) AS total_balance,
SUM(CASE WHEN Exited = 1 THEN Balance ELSE 0 END) AS balance_lost,
ROUND(
  SUM(CASE WHEN Exited = 1 THEN Balance ELSE 0 END) / SUM(Balance) * 100, 2) AS
  loss_percentage,
ROUND(SUM(Exited)*100.0/COUNT(*),2) AS churn_rate
FROM bank_churn_clean
GROUP BY Geography
ORDER BY balance_lost DESC;
```